

ENGINEERING DESIGN REPORT

132 kV Transmission Line Tower Foundation Design

Project Overview

This report presents the design methodology for an RCC foundation supporting a 132 kV double-circuit transmission line tower. Design complies with IS 456:2000, IS 800:2007, IS 875 (Part 3):2015, IS 1893:2016 and IEC 60826.

Site Information

Location: Chennai, Tamil Nadu
Soil: Medium Dense Sand
Safe Bearing Capacity: 250 kN/m²
Water Table: 3.2 m
Seismic Zone: III
Basic Wind Speed: 50 m/s

Tower Details

Tower Type: Double Circuit Suspension
Voltage: 132 kV
Height: 36 m
Base Width: 7.5 m

Design Loads

Dead Load: 320 kN
Live Load: 60 kN
Wind Load: 180 kN
Total Vertical Load: 560 kN
Uplift: 210 kN
Longitudinal: 95 kN
Transverse: 140 kN

Foundation Design

Type: Isolated RCC Pad Foundation
Dimensions: 3.2 × 3.2 × 2.8 m
Concrete Grade: M30
Steel Grade: Fe500

Bearing Pressure Check

Area = 10.24 m²
Bearing Pressure = 54.7 kN/m²
Allowable SBC = 250 kN/m²
Status: SAFE

Reinforcement

Bottom: 20 mm @150 mm c/c
Top: 16 mm @200 mm c/c

Development Length: 920 mm
Lap Length: 1000 mm

Quantities

Concrete Volume: 28.67 m³
Steel Consumption: 118 kg/m³
Total Reinforcement: 3383 kg

Factor of Safety

Sliding: 2.35 (PASS)
Overturning: 2.92 (PASS)
Bearing Failure: 4.10 (PASS)

Inspection Checklist

Concrete tested; Reinforcement inspected; Anchor bolts aligned; Excavation approved; PCC completed; Earthing installed; Curing completed.

Maintenance

Annual inspection; Monitor settlement; Tighten anchor bolts; Repair cracks >0.3 mm; Corrosion inspection every 2 years.

Conclusion

The foundation satisfies structural safety requirements. Bearing pressure is below allowable limits and all factors of safety exceed minimum code requirements.